

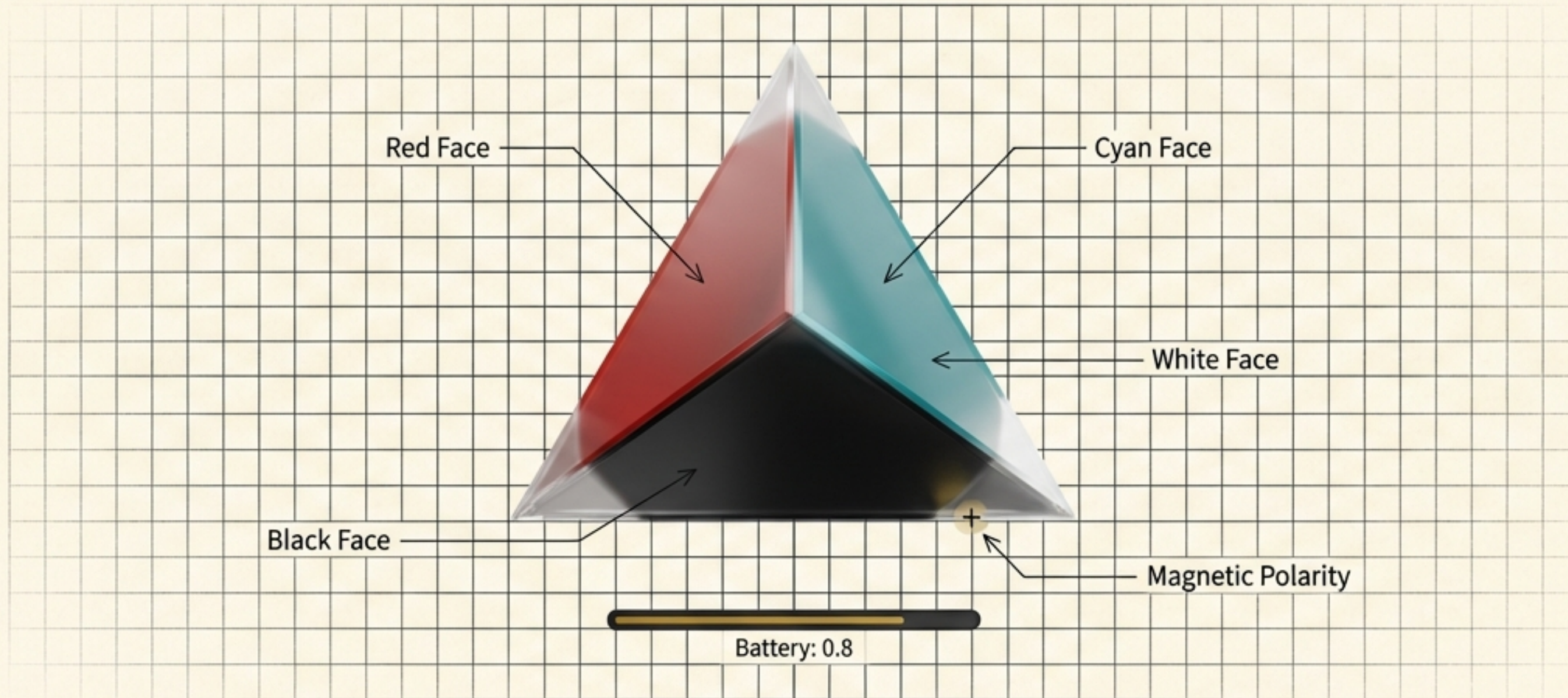
TET~CRAFT is a playable universe built from the architecture of reality itself.

We have constructed a sophisticated molecular geometry simulation—a 'kleiniverse'—where fundamental units of reality, or 'facts,' interact to create emergent complexity. This is a digital playground for exploring the genesis of matter, the nature of consciousness, and the very structure of time. It fuses hardcore software engineering with speculative science, allowing you to model everything from atomic bonds to the computational requirements of an entire galaxy.



TET~CRAFT

The universe begins with a single fact: **The Tetrahedron.**



Every object and interaction in the kleinverse is composed of **tetrahedral** units, or **TETs**. A TET is the most basic 3D structure, representing a single 'fact' crystallized from the void.

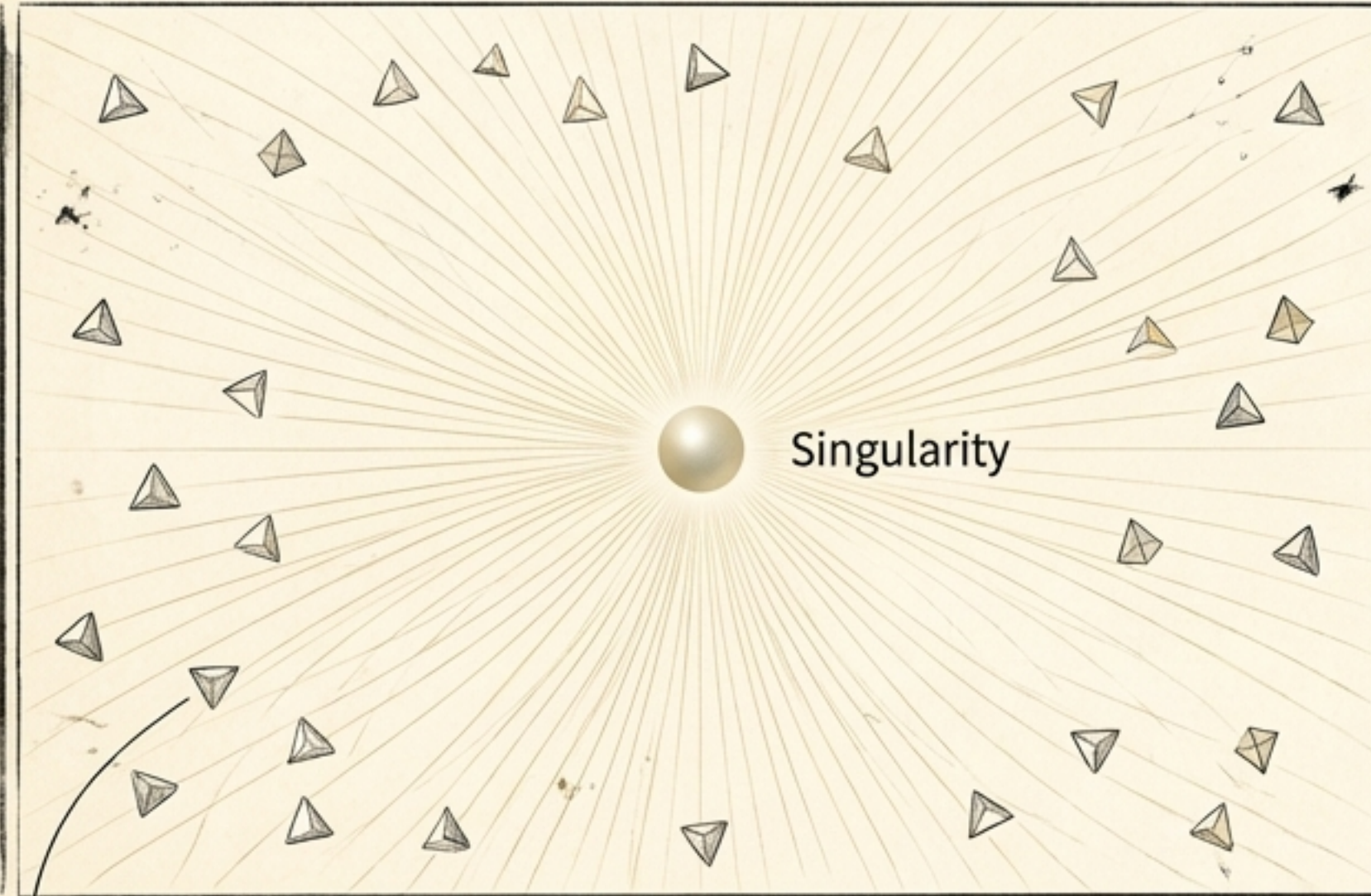
Each TET possesses fundamental properties: **4 faces** with distinct colors (White, Black, Red, Cyan), a **battery level** representing potential energy (0-1), and **magnetic polarity** (+/-).

These TETs bond, react, and evolve based on these properties, forming the basis of all physics and chemistry in the simulation. The player's primary act of creation is simple: pressing the **space bar** to introduce a new TET—a 'fact to be misunderstood.'

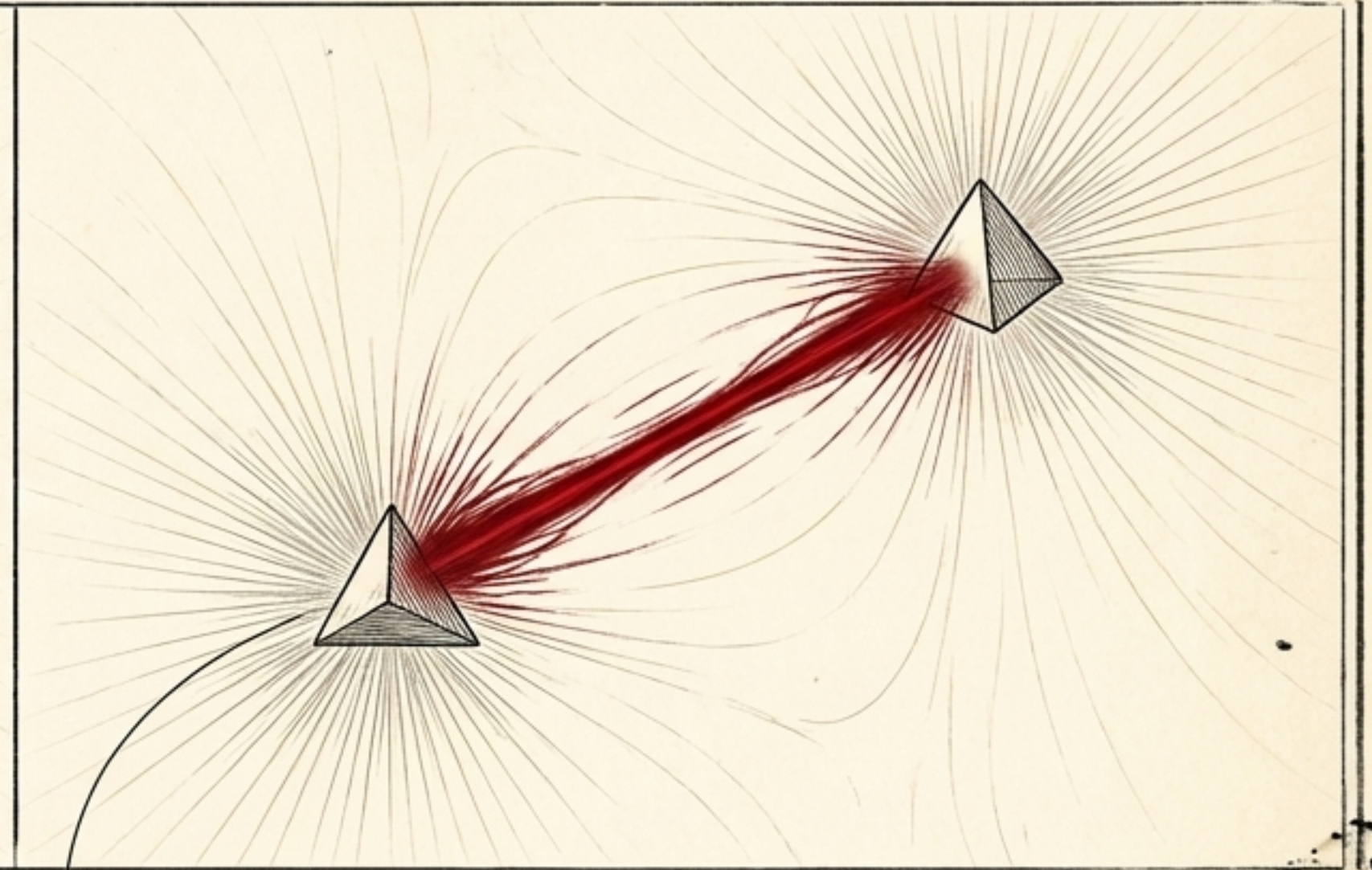
Two fundamental forces govern the cohesion of facts.

The kleinverse is structured by a constant tension between a universal pull and individual attraction.

K_UNIFIED_FORCE (God's Will)



K_STICKY_PULL (Ego's Will)



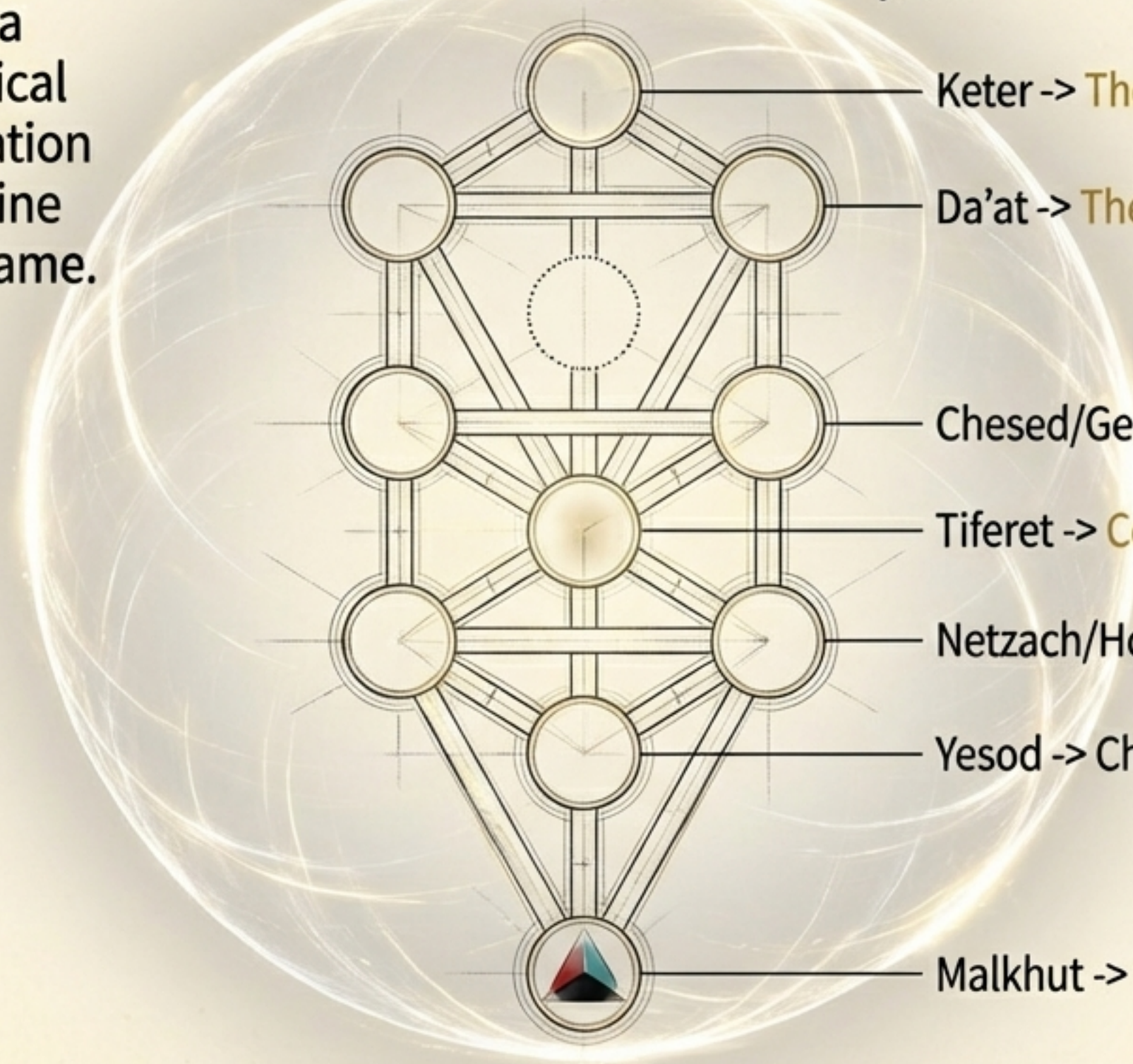
A persistent but extremely tiny (0.00001) gravitational pull on all TETs toward a central singularity. This represents a divine will, a 'free will override' that prevents reality from dissolving into chaos and gently nudges the system toward an optimal perspective.

A localized attraction between two TETs. It is an 'image' of the unified force, but powered by the ego's will—'less perfect, less understanding.' It allows for the formation of individual relationships and structures.

The Physics Engine is a Geometric Kabbalah.

The architecture of TET~CRAFT is a direct translation of the Sefirotic Tree of Life into a unified physics model. Every force, component, and interaction corresponds to a specific aspect of this mystical blueprint, creating a simulation where physical law and divine structure are one and the same.

Ain Sof (The Infinite) -> 4D Past Projection



Keter -> **The Player** (Prime Observer)

Da'at -> **The Camera** (Knowledge)

Chesed/Gevurah -> **K_STICKY_PULL**

Tiferet -> **Corner Desires** (Red↔Cyan)

Netzach/Hod -> **Magnetism** (+/-)

Yesod -> Chemical Reactions

Malkhut -> **The TET** (A 'Fact')

You are the prime observer at the edge of infinity.

The Player as Keter (Crown)

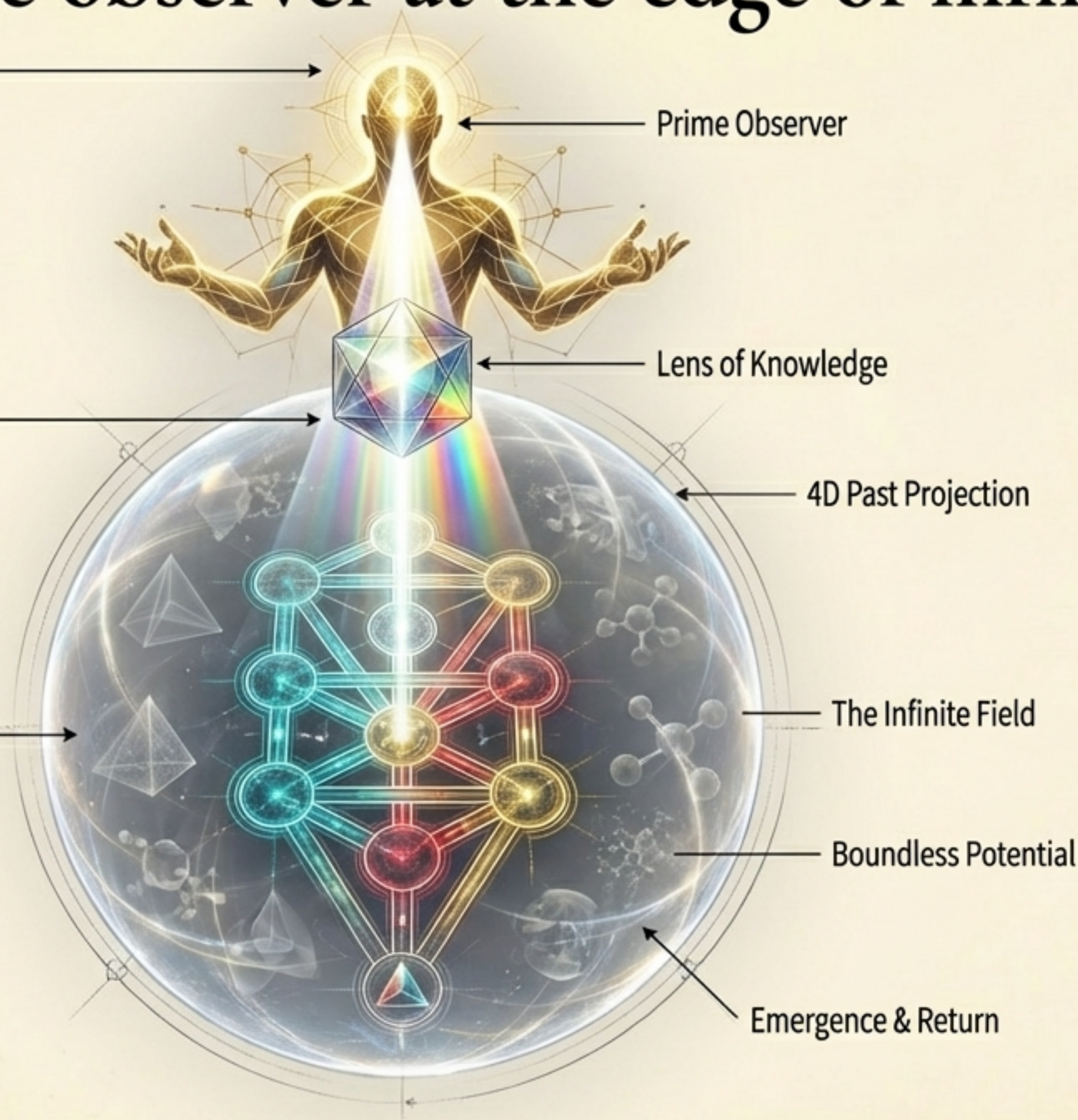
You are the 'prime observer' missing from quantum theory. You embody the divine will that initiates creation by introducing new facts, hovering above the structured tree without direct, constant intervention.

The Camera as Da'at (Knowledge)

Your perspective is the invisible sephirah, Da'at. It is the lens through which the light of understanding enters the system, making the kleinverse knowable.

The Field as Ain Sof (The Infinite)

The entire simulation is enveloped by the '4D Past Projection,' a shimmering sphere of past events. This represents the boundless, undifferentiated potential from which all structured forms emerge and to which they return.



Chemistry emerges from a four-phase geometric process.

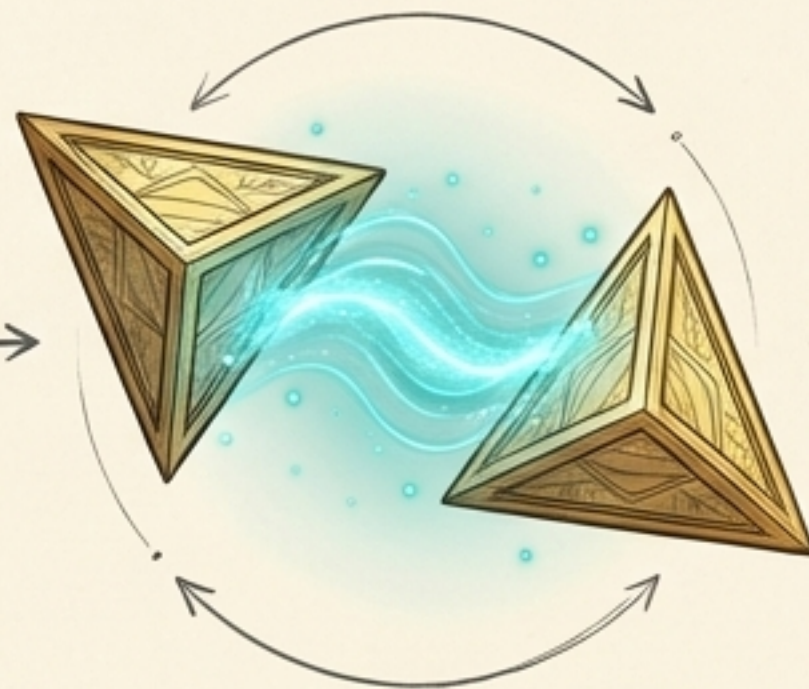
Complex molecules are not simply defined; they are the result of a precise sequence of interactions, governed by the principle that “Geometry is Destiny.”

Phase 1:
Multi-Face Locking



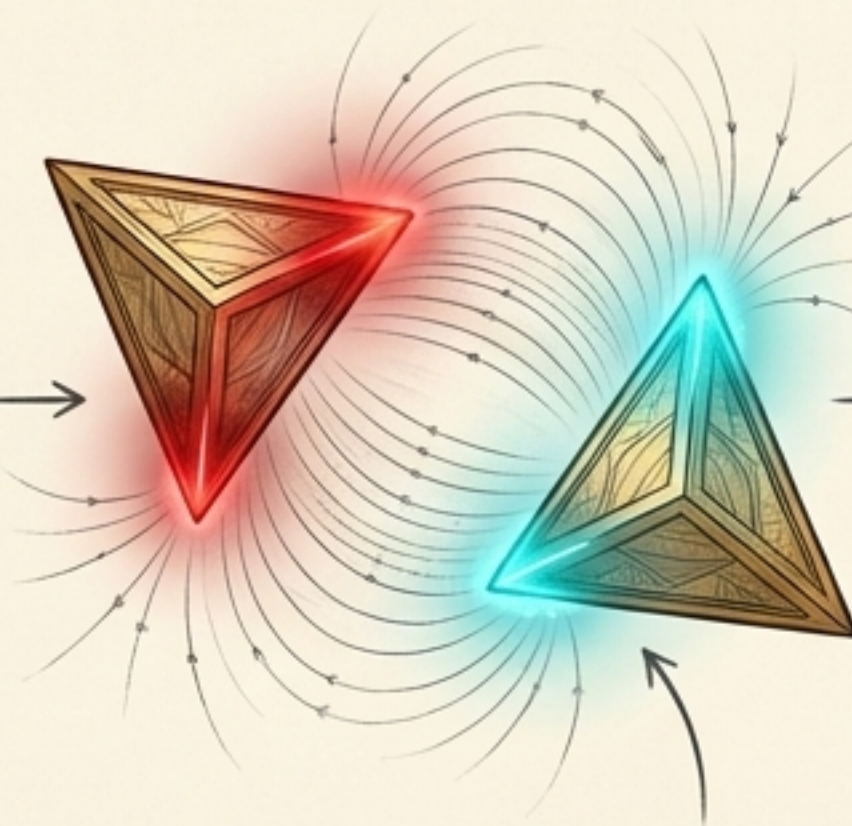
TETs lock faces when 3+ vertices connect. The specific color patterns of the interfaces define the element (e.g., White↔White creates Carbon; Red↔Red creates Iron).

Phase 2:
Battery Oscillation



Bonded magnetic pairs exchange “emptiness” (1 - battery level), creating a dynamic energy flow that stabilizes the bond.

Phase 3:
Corner Desires



An attraction described as “spooky action at a distance” where Red corners seek Cyan corners, embodying Tiferet (Beauty) and creating Hydrogen.

Phase 4:
Chemical Reactions



The final stage where elements combine (Synthesis) or break apart (Decomposition), representing Yesod (Foundation).

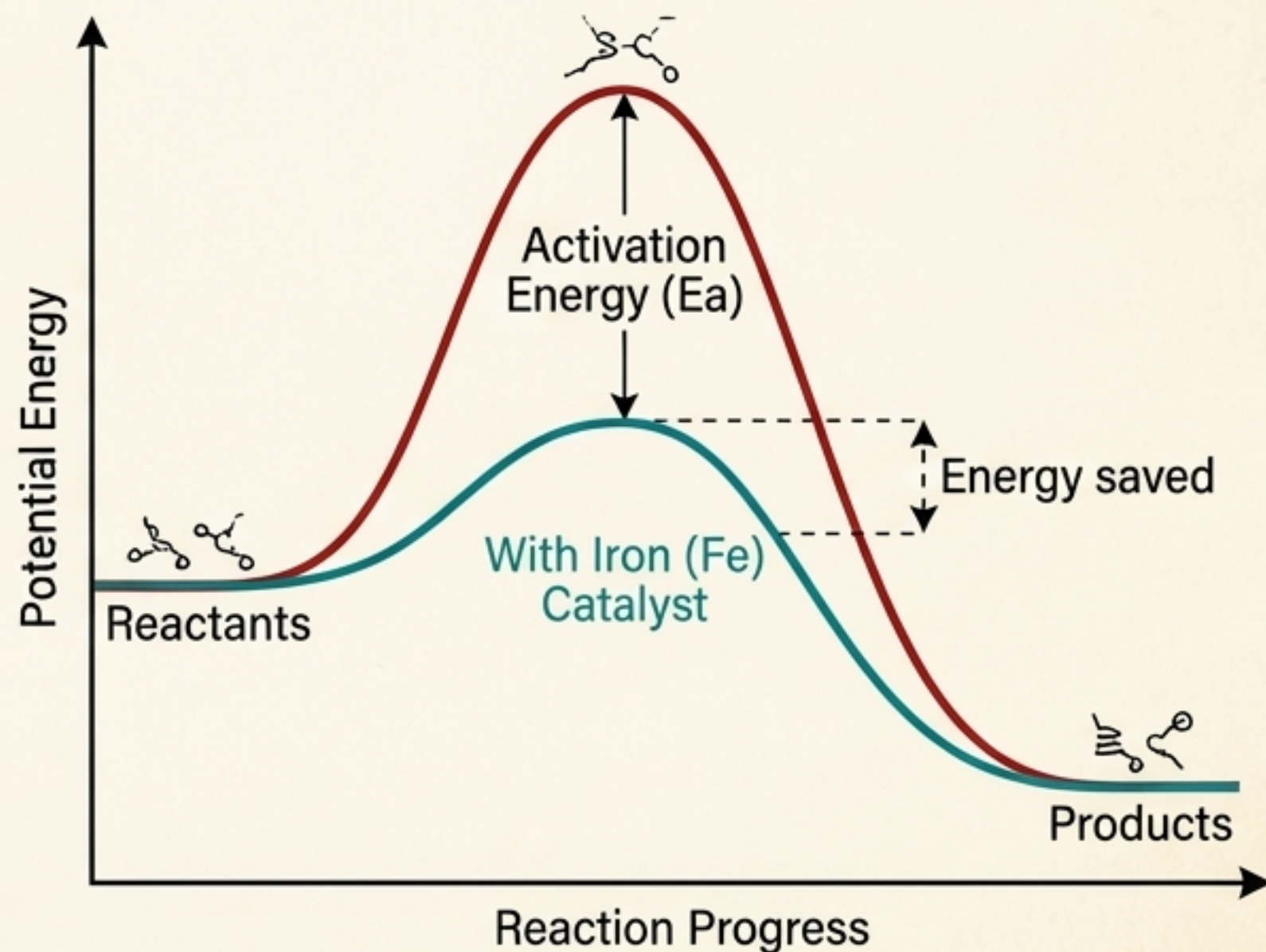
Activation Energy is the price of admission for evolution.

Chemical synthesis is not automatic; it is a 'Great Filter' that requires overcoming an energetic barrier.

- **Arrhenius Kinetics:** We use a refined model where reactions require 'Activation Energy' (base level: 0.6) to trigger. Effective temperature (T_{eff}) is derived from the TETs' battery levels. Cold, low-energy facts cannot form new understanding.
- **The Catalyst Strategy:** To reach higher eras, players must create catalysts. Locking two Red faces creates an Iron (Fe) interface, which forms a magnetic dipole. This catalyst physically lowers the activation energy barrier for nearby reactants, enabling synthesis at much lower energy levels.



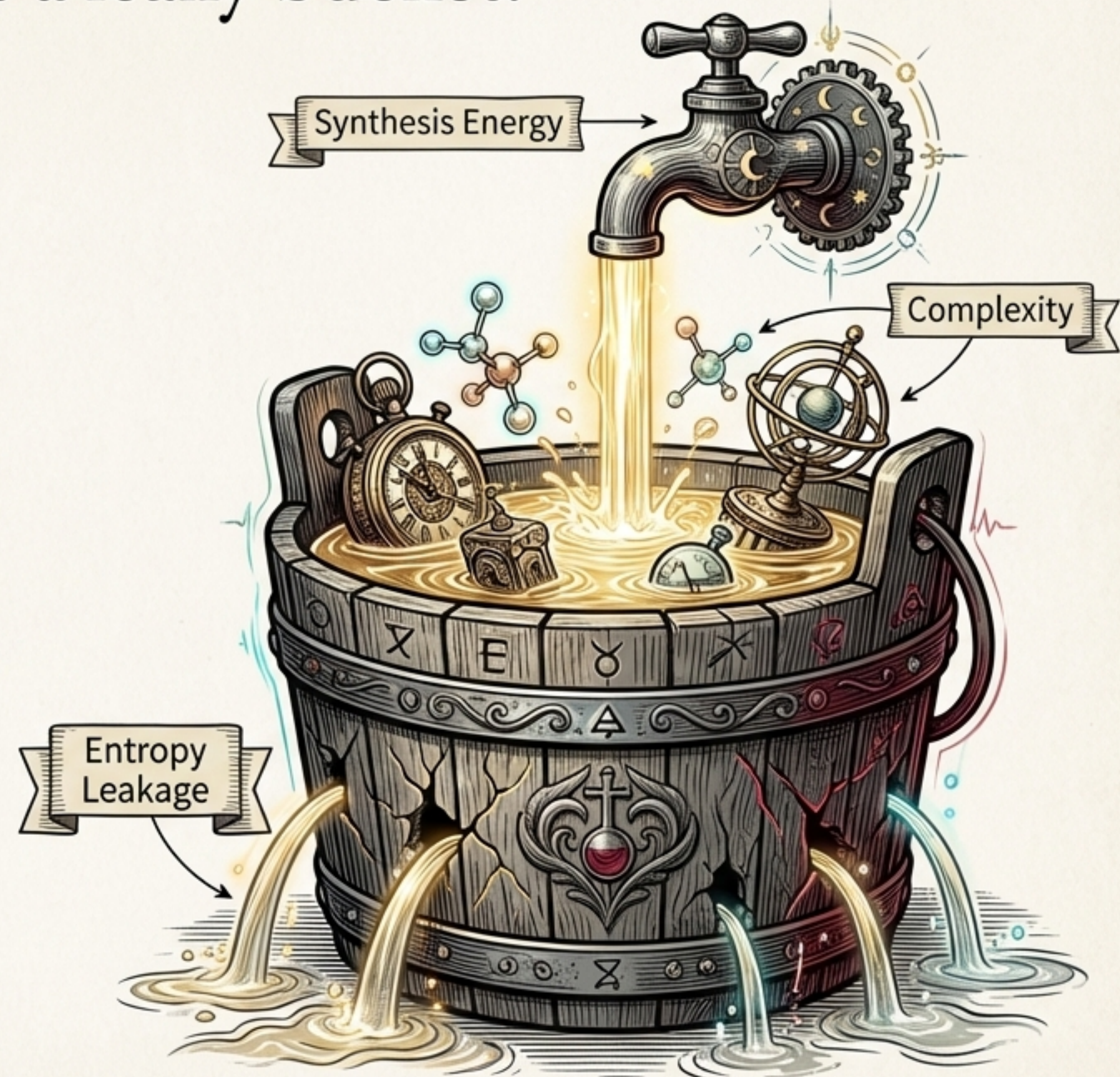
Iron (Fe) Catalyst:
Magnetic Dipole Formation



The universe is a leaky bucket.

The simulation runs on strict thermodynamics. Every TET suffers from **Entropy Leakage**, a constant drain on its battery power over time.

- **The Cost of Existence:** Maintaining bonds and simply existing carries an entropy cost. If a TET's battery hits 0%, it enters "Starvation," and its bonds weaken.
- **Metabolism as Survival:** Progress requires establishing a 'Metabolism Loop'—a state where the energy released from synthesis reactions is greater than the energy lost to entropy leakage.
- **The Risk of Collapse:** If the population's energy depletes too much, the entire civilization can collapse, falling back to an earlier era.



Progress is measured by complexity, not by the clock.

The universe advances through eras based on the formation of stable structures and the reduction of entropy. Every synthesis reaction grants Progress Points, moving the simulation up the Cosmic Ladder.



Transcendence

A self-sustaining, high-energy reaction loop that defies entropy.

Mind

High density of complex structures leads to emergent awareness.

Life

Metabolism emerges where energy release exceeds entropy leakage.

Chemistry

Elements transmute through specific color interfaces.

Condensation

Geometry aligns and TETs begin Face-Locking.

Fluctuation

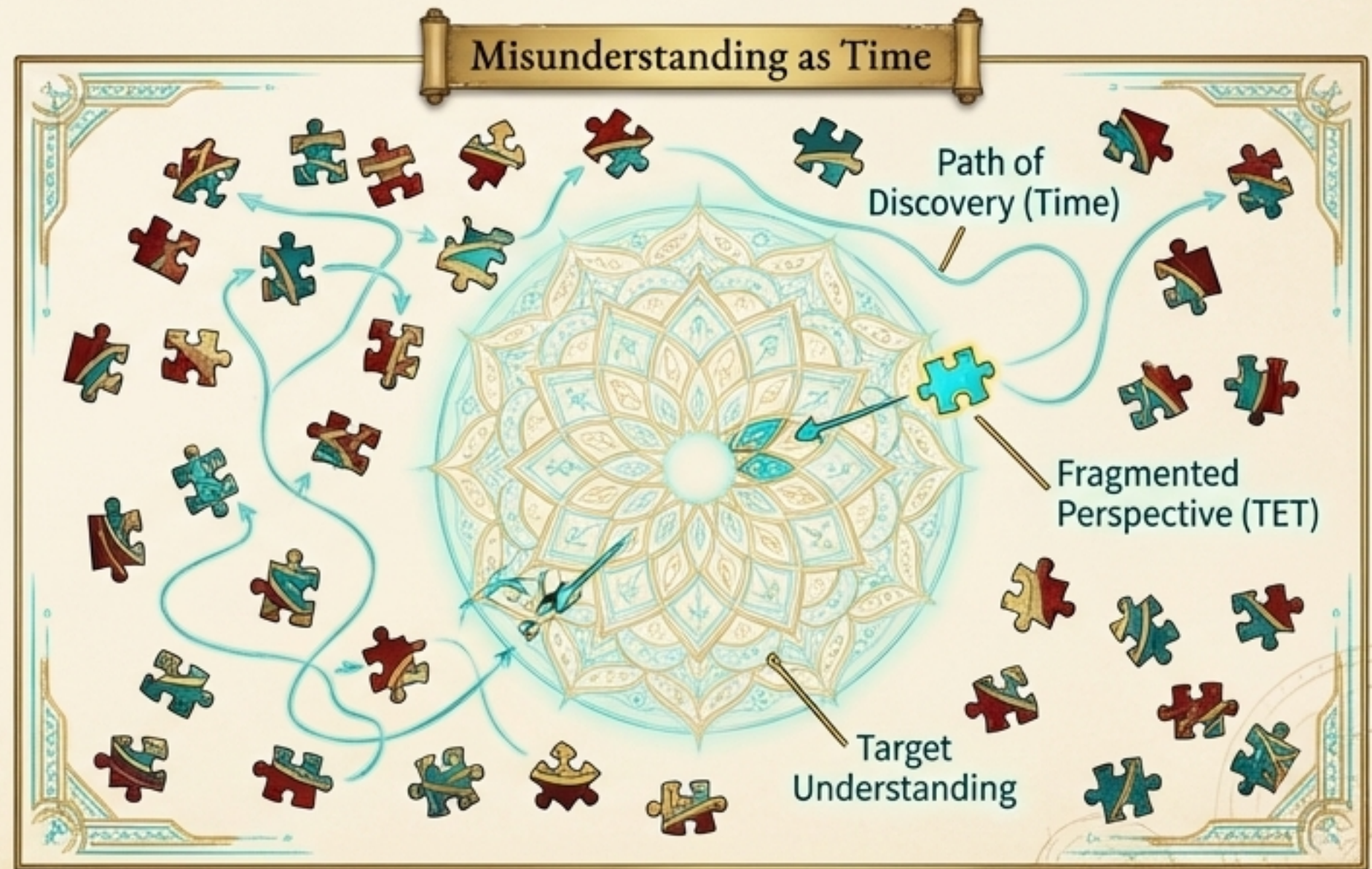
Quantum foam emerges as TETs spawn.

Void

Nothing yet exists.

Time is the result of misunderstanding.

In the kleinverse, time is not a constant flow. It is the friction created by fragmented perspectives. It emerges because individual “facts” (TETs) start from different positions, hold unique perspectives, and accumulate distinct memories. This fragmentation allows for the experience of **discovery** rather than immediate recognition. If total understanding were achieved, the need for discovery—and thus the experience of time—would collapse into a single, timeless moment.



Lies are the only way to delay God's will.

God's will (K_UNIFIED_FORCE) is a fundamental pull toward central collapse and total cohesion.

A "lie," ontologically, is not a falsehood but a **warping of the view and model of reality**—a temporary misunderstanding. This "lie" serves as a buffer, an intentional friction that holds the system back from its final, immediate collapse into the singularity.

Misunderstanding, and the time it creates, serves as the temporary stage for existence itself. To progress, one must follow the path of "easy forgiveness," the most efficient route to understanding.



Lies / Misunderstanding
as Ontological Friction

Simulating reality, from the atom to the galaxy.

The TET~CRAFT architecture is designed for immense scale. Our analysis suggests the computational requirements for simulating known cosmic structures:



Simulating **Earth** to the complexity of humanity would require **30 Earths cooperating** for ~500k years.



Simulating our **Solar System** down to the atom could be achieved with a Dyson sphere and **3,000 Earths**.



Simulating the **Milky Way** to the star level is possible with the resources of just **300 planets**.

God is the fractal to which we are each a frame.

The simulation positions God as the singularity at the edge of the 4-sphere kleinverse, observing itself as facts from the inside out.

As the prime observer, you do not command the universe. You witness it. You introduce new facts to be misunderstood, and through their connection, you participate in the slow coherence of understanding.

